

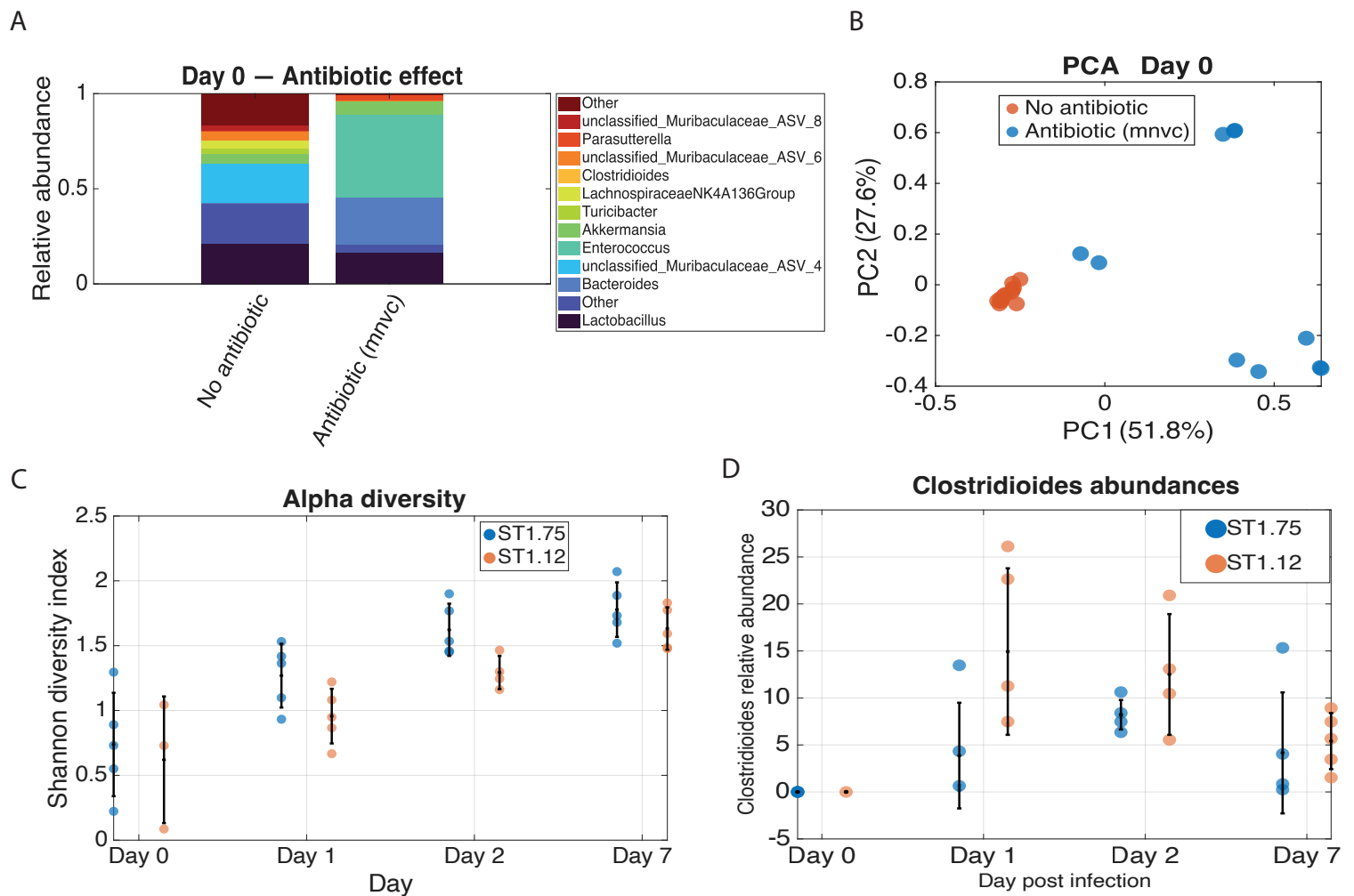


Figure 3: Precolonization with ST1-75 does not substantially alter remnant microbiota composition. (A) Relative abundance of major bacterial taxa in fecal samples at Day 0 from mice with no antibiotic treatment and mice treated with the MVN antibiotic cocktail, assessed by 16s amplicon sequencing. Each bar represents a pooled community composition showing the effect of antibiotic-induced dysbiosis prior to *C. difficile* colonization. (B) Principal coordinates analysis (PCA) of Bray-Curtis dissimilarity at Day 0 for no-antibiotic (orange) and antibiotic-treated (blue) mice. PC1 explains 51.8% and PC2 explains 27.0% of the variance. The two groups are compositionally distinct at baseline, confirming the dysbiotic state induced by MVN treatment. (C) Shannon diversity index over time (Day 0–7 post-colonization) for mice colonized with ST1-75 (blue) or ST1-12 (orange). No significant differences in alpha diversity were observed between the two groups at any timepoint (Day 1: $p = 0.095$; Day 2: $p = 0.063$; Day 7: $p = 0.310$). Each point represents an individual mouse; bars indicate the mean. (D) Relative abundance of the *Clostridioides* genus over time (Day 0–7) for ST1-75 (blue) and ST1-12 (orange) colonized mice. Each point represents an individual mouse; bars indicate the mean.